GEYI YANG

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Research Interests

LLM-based agents, GUI agents, self-improving agentic systems, harness optimization, and retrieval-augmented generation (RAG).

Education

- **The Chinese University of Hong Kong, Shenzhen** *Sep. 2023 – Present*
  B.Eng. in Computer Engineering Shenzhen, China
  - GPA: 3.73/4.00.
  - Selected coursework: Artificial Intelligence, Machine Learning, Optimization, Data Structures, Operating Systems, and Internet Architecture and Protocol.
- **University of California, Berkeley** *Jan. 2026 – Jun. 2026*
  GLOBE Program Visiting Student in Computer Science Berkeley, CA, USA

Honors and Awards

COMAP Mathematical Contest in Modeling (MCM), Finalist (F Award; top 1%) 2026
China International College Students' Innovation Competition (CICSIC), Bronze Award 2025
Dean's List, CUHK-Shenzhen 2023–2024, 2024–2025, 2025–2026
Academic Performance Scholarship Awardee, CUHK-Shenzhen 2023–2024

Preprints

[1] **Geyi Yang**, Zikun Qu, Xiang Li, Zhiyong Wang, Min Zhang, Shipei Zeng, and Zhongxiang Dai. "GUI-HARVEST: Self-Improving GUI Agents through Evidence-Driven Harness Evolution." *arXiv preprint arXiv:2610.00948*, 2026. Under review at ICLR 2027. [arXiv]

Research Experience

- **GUI-HARVEST: Self-Improving GUI Agents through Evidence-Driven Harness Evolution** *Jul. 2026 – Present*
  Research Assistant, Shenzhen Research Institute of Big Data | *Advisor: Zhongxiang Dai* GitHub
  - First-authored GUI-HARVEST (under review at ICLR 2027), an automatic harness optimizer that enables self-improving GUI agents with frozen backbone models by converting repeated multimodal executions into validated runtime code changes.
  - Designed an evidence-driven loop that aligns actions with before-and-after screenshots, compares repeated runs, clusters recurring cross-task failures, and promotes edits only after score and predicted-behavior checks.
  - Achieved held-out gains across six open and proprietary backbones on OSWorld-Verified; Qwen3-VL-32B-Instruct gained 12.33 points on the full suite, while frozen-harness transfer improved GPT-5 by 13.87 points on WindowsAgentArena.
- **Meme-Qwen (7B-Instruct)** *Jan. 2026*
  Chinese Internet-Meme Conversational Language Model Model | Dataset
  - Independently built and open-sourced, to our knowledge, the first Chinese internet-meme conversational SFT dataset and the first open model of this type: collected 30,000+ post-comment pairs with Playwright and released 8,680 ChatML examples.
  - Fine-tuned Qwen2.5-7B on a single A100 (40GB) with LoRA and DPO; improved meme-use accuracy by 32%, with human and LLM preference rates of 82%/91%/68% for humor, accuracy, and naturalness.

Internships

- **Cloud Computing Center, Chinese Academy of Sciences** *Mar. 2025 – Jun. 2025*
  Domain-Specific Agentic RAG for Mining-Blasting QA Dongguan, China
  - Built and optimized an Agentic RAG system for retrieving and answering questions over mining-blasting technical documents, owning the end-to-end retrieval and reasoning pipeline.
  - Architected a hierarchical LangGraph workflow for memory, query planning, answer synthesis, and a ReAct subgraph for tool use and multi-step reasoning; added human clarification for ambiguous queries.
  - Developed Docling-based parsing and parent-child hybrid retrieval in Qdrant, combining BGE-M3, BM25, and BGE-Reranker-v2-m3; achieved 81% Recall@5 and 78% answer accuracy on 300+ QA cases.
- **Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)** *Sep. 2023 – Jun. 2024*
  Multimodal Digital Human Synthesis for Emotional Interaction Shenzhen, China
  - Developed an end-to-end "Digital Relative" pipeline from family photos and recordings, integrating MockingBird voice cloning, SadTalker lip-sync and facial animation, and ChatGPT-powered dialogue.
  - Piloted the system with family users in 500+ interactions: 83% reported a familiar emotional connection and 20% reported a strong emotional response.

Products

- **StashDock: A Visual Context Library for AI** *Oct. 2026 – Present*
  Creator and Full-Stack Developer Website | GitHub
  - Created and launched a visual context library that lets users save webpages, files, prompts, and rules in one place, then select and send them to AI conversations without repeatedly searching bookmarks, folders, and past chats.
  - Designed the workflow from collection to reuse: capture pages with a Chrome extension or import bookmarks/files, organize references into tagged context sets, and copy or drag them into any AI chat or add them directly through a ChatGPT sidebar.
  - Independently owned product design, UI/UX, and end-to-end development; shipped open-source Windows/macOS apps with local storage and MCP/Skills integration that lets AI find relevant saved references and preserve useful prompts and rules for future tasks.

Selected Projects

- **MCM 2026: Dynamic Hybrid Lunar Logistics Optimization** *2026*
  Modeling and Programming Lead Finalist (F Award; top 1%)
  - Developed a dynamic Earth-to-Moon logistics model integrating space elevators, reusable rockets, and ISRU maturation; formulated time-cost objectives and operational constraints.
  - Used Differential Evolution (1,301 evaluations), Monte Carlo analysis (5,000 simulations), and sensitivity analysis to identify a robust three-phase transition strategy; awarded Finalist (top 1%).
- **Telos: AI-Generated, Ready-to-Learn Courses** *Aug. 2026 – Present*
  Independent Full-Stack Developer GitHub
  - Building a goal-conditioned education product that turns user materials and cross-platform resources into structured, multimodal, ready-to-learn courses rather than isolated answers or learning plans.
  - Designed an agentic pipeline for retrieval, deduplication, quality assessment, sequencing, and embedding source text, images, and videos in a unified learning workspace.
- **Voice Turing Test: WeChat Mini Program and Research Data Platform** *May 2025 – Aug. 2025*
  Full-Stack Development Lead
  - Independently led registration, UI/UX, full-stack development, testing, and deployment of a gamified Human-vs-AI speech challenge used as the data-collection platform for *Human or Machine?* (ICLR 2026).
  - Built the native WeChat front end and Tencent Cloud backend with bilingual UI, real-time scoring, achievements, leaderboards, and referral rewards; reached 1,000+ users, 300+ DAU, and 5,000+ valid records in the first month.

Teaching Experience

- **The Chinese University of Hong Kong, Shenzhen** *Sep. 2025 – Dec. 2025*
  Teaching Assistant – MAT1001 Calculus I Shenzhen, China
  - Served as a teaching assistant for the first-year calculus course MAT1001, supporting tutorials and student learning.

Skills and Languages

Programming & Systems: Python, C++, JavaScript, TypeScript, Shell, FastAPI, React, SQLite, Git, Linux.
Machine Learning & Agents: PyTorch, Transformers, LoRA/PEFT, TRL (DPO), RAG, LangChain, LangGraph, GUI agents, Qdrant, MCP.
Languages: English (TOEFL 111; GRE 332).